

HTML Revision Notes

1. Introduction to HTML

- **HTML (HyperText Markup Language)** is used to structure web pages.
- index.html is usually treated as the homepage of a website.
- HTML documents are made up of **elements**.
- **HTML Element = Opening Tag + Content + Closing Tag**

Example:

```
<p>Hello World</p>
```

2. Basic HTML Structure

<html>

- Root element of every HTML document.
- Contains all other HTML elements.

<head>

- Stores metadata and information about the page.
- Content inside <head> is not displayed directly on the webpage.

<body>

- Contains all visible content displayed by the browser.
-

3. Comments

- Used to add notes inside code.
- Ignored by the browser.

```
<!-- This is a comment -->
```

4. Attributes

- Provide additional information to HTML elements.
- Written inside the opening tag.

Example:

```
<a href="https://google.com">Google</a>
```

Here, href is an attribute.

5. Links (<a> Tag)

The anchor tag is used to create hyperlinks.

Absolute Links

- Point to external websites.

Example:

```
<a href="https://www.google.com">Google</a>
```

Relative Links

- Point to pages within the same project.

Example:

```
<a href="about.html">About</a>
```

Open Link in New Tab

```
<a href="https://google.com" target="_blank">Google</a>
```

6. Images (Tag)

Used to display images on a webpage.

Example:

```

```

Image Sizing

- Set either width or height.
- The other dimension adjusts automatically to maintain aspect ratio.

Example:

```

```

7. Text Formatting Tags

Tag	Purpose
	Bold text
<i>	Italic text
<u>	Underlined text
<s>	Strikethrough text
<big>	Larger text
<small>	Smaller text

Example:

```
<b>Bold Text</b>
```

8. Line Break and Horizontal Rule

Line Break

```
<br>
```

Moves content to the next line.

Horizontal Rule

```
<hr>
```

Creates a horizontal separator line.

9. Subscript and Superscript

Subscript

Used for formulas like H₂O.

```
H<sub>2</sub>O
```

Superscript

Used for powers like x^2 .

`x<sup>2</sup>`

10. Preformatted Text (`<pre>`)

- Preserves spaces and line breaks exactly as written.

Example:

```
<pre>
Hello
  World
</pre>
```

11. Semantic vs Non-Semantic Tags

Semantic Tags

Clearly describe their purpose.

Examples:

```
<header>
<footer>
<main>
<section>
<article>
<aside>
```

Benefits:

- Better readability
- Improved accessibility
- Better SEO

Non-Semantic Tags

Do not describe their content.

Examples:

```
<div>
<span>
```

12. Page Layout Structure

Common page structure:

<header>
<main>
<footer>

Inside <main>

<section>

Groups related content.

<article>

Represents independent content such as blog posts or articles.

<aside>

Contains content related to the main content, such as advertisements or sidebars.

13. Containers

<div>

- Block-level container.
- Takes full available width.
- Used to group elements.

- Inline container.
 - Takes only the required width.
 - Used for styling or grouping small portions of text.
-

14. Lists

Unordered List ()

Displays items using bullet points.

```
<ul>
  <li>Apple</li>
  <li>Mango</li>
</ul>
```

Ordered List ()

Displays items using numbers.

```
<ol>
  <li>First</li>
  <li>Second</li>
</ol>
```

15. Tables

Table Structure

Tag	Purpose
<table>	Creates a table
<tr>	Table row
<th>	Table header
<td>	Table data

Example:

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Ali</td>
    <td>20</td>
  </tr>
</table>
```

Table Sections

<thead>

Contains table headers.

<tbody>

Contains table data.

Merge Columns

```
<td colspan="2">Merged Cell</td>
```

colspan merges multiple columns into one.

16. Forms

Forms are used to collect user input.

Common Form Elements

Label

```
<label>Name</label>
```

Input Types

```
<input type="text">  
<input type="email">  
<input type="password">  
<input type="radio">  
<input type="checkbox">
```

Select Dropdown

```
<select>  
  <option>Option 1</option>  
</select>
```

17. Class and ID

Class

- Can be used by multiple elements.
- Used for grouping similar elements.

Example:

```
<p class="info">Text</p>
```

ID

- Must be unique on a page.

- Identifies a single element.

Example:

```
<p id="heading">Text</p>
```

Quick Difference

Class	ID
Reusable	Unique
Multiple elements can share it	Only one element should use it
Selected using . in CSS	Selected using # in CSS